



Super Mario's record

Nintendo's new Super Mario game on iOS sets a new record with 40million downloads in 4 days. <http://dgit.in/SuperMR>



Prisma social feel

Prisma, the popular photo editing application launches their new social feed. Learn more here: <http://dgit.in/PrismaSF>

has hence kept the precise postulates under research, but we do seem to have a fair idea.

Superluminous supernovae typically last for over six months, as compared to the duration of 2-3 weeks for a standard supernova. The reactions, too, are different in nature, and far more titanic than the standard incident of the death of a



Now you know what happens when two stars collide

star. Multiple causing factors have been identified, of which we will talk about in the next section. These are also believed now to be a significant source of gamma ray bursts in the cosmos, although concrete outtakes are yet to be derived of it.

As against a typical supernova, the superluminous supernovae are said to feature two sections in the stellar explosion. Before the major explosion that sustains over months and shines consistently brighter than its surrounding galaxy for a much longer span, there is a smaller, initial explosion - a starter ignition of sorts - that leads to a subsequent dip spanning over a few days. It is following this that there is a surge in the energy and light emission of the superluminous supernova.

This entire procedure is what makes this phenomenon rare, extremely volatile and unusual. There are even more titanic explosions across our galaxies far, far away (not akin the Death Star, of course) that shine even brighter. For instance, an explosion observed by the All-Sky Automated Survey for Supernovae (ASASSN) was from a source that takes light 3.8 billion years to reach, and was reportedly twice as bright as the largest known superluminous supernova.

The reason of this explosion is now being attributed to the impact of a gar-

gantuan black hole that literally ripped a massive star apart, to create light and energy that is 570 billion times higher than the Sun's peak light and energy radiation. While this will indeed go down as the death of a star, and technically fall under a superluminous supernova, our speculations would say that this is certainly an infrequent occurrence. If



A giant star was torn apart to create the brightest galactic observation yet

not, though, then we have forces out there that are far greater than what we have known of.

This, to sum up, is a superluminous supernova. The violent death of a massive star, catalysed by anomalous elements to form fireballs of energy that is way beyond standard comprehension. How do they occur, though?

This is how they are formed

Before we begin, it is essential to understand that the causal factors of a superluminous supernova are highly unusual, and as such, have been pegged as possibilities. Not always can the exact reason of the formation of a superluminous supernova be understood in accordance to their type, simply because of their highly complicated structure and massive parameters of the explosions.

A supernova typically forms when the core of a star collapses to give rise to a gravitational pullback of energy. When this happens, there are multiple times when the core is too heavy, and the core's explosive propulsion is not enough to eject the outer shells of the star. This leads to a silent death of a star that ends up forming black holes.

Supernovae are formed only when the core of the star weighs between 5 and 15 times the mass of our star, the

Sun. When the gravitational collapse of such a star begins, a faint supernova is produced when the star has slow rotational rate. The slow speed ejects smaller jets to outer orbits over short distances, thereby forming the essence of a standard, or in our case, a faint supernova.

Considering the chemical constituency of the dying stars in our examples are constant, a superluminous supernova will occur only when the star has very fast rotational speeds. This leads to light and energy being spewed far beyond the outer shells as the core falls through in gravitational collapse, and these jets of light and energy are relative in magnitude to the rotational speed. This is what makes a more 'usual' superluminous supernova, which also leads to long outbursts of gamma rays.

However, there are further questions within the primary postulate of stellar explosions leading to superluminous supernovae. While the mass and rotational speeds do contribute to the formation, the chemical constituency is also a major deciding factor on the feasibility of titanic explosions. Typically, a star that ends up in a superluminous supernova has higher metallicity alongside the



The force is literally quite strong here

standard Hydrogen-Helium composition, which further leads to very high surface temperatures and luminosity at death. It is this that makes them shine brighter and at significantly higher temperatures for sustained durations, in which it surpasses the energy given by a standard supernova.

While the previous model explains superluminous supernovae with high gamma ray outbursts, the other explanation is of stellar deaths that do not